

Exploring place names through new computational tools: Examples from India

Lennart Chevallier & Søren Wichmann

Kiel University

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Structure of the talk

- Place names: A brief introduction
- The GeoNames.org resource
- Nuts and bolts of our R package
- Toponymic studies in India
- Examples from India: Konkan region
- Conclusions and outlook

Why study the distribution of toponyms?

- Place names can reveal the language of people occupying a certain space
- More often than not, these place names will not be replaced even when the ethnic profile changes, although their shape can change
- Most studies of toponymics are about single names or individual names of a certain region, while distributional studies are rarer (e.g., next to no distributional maps in the most recent textbook we know of¹)

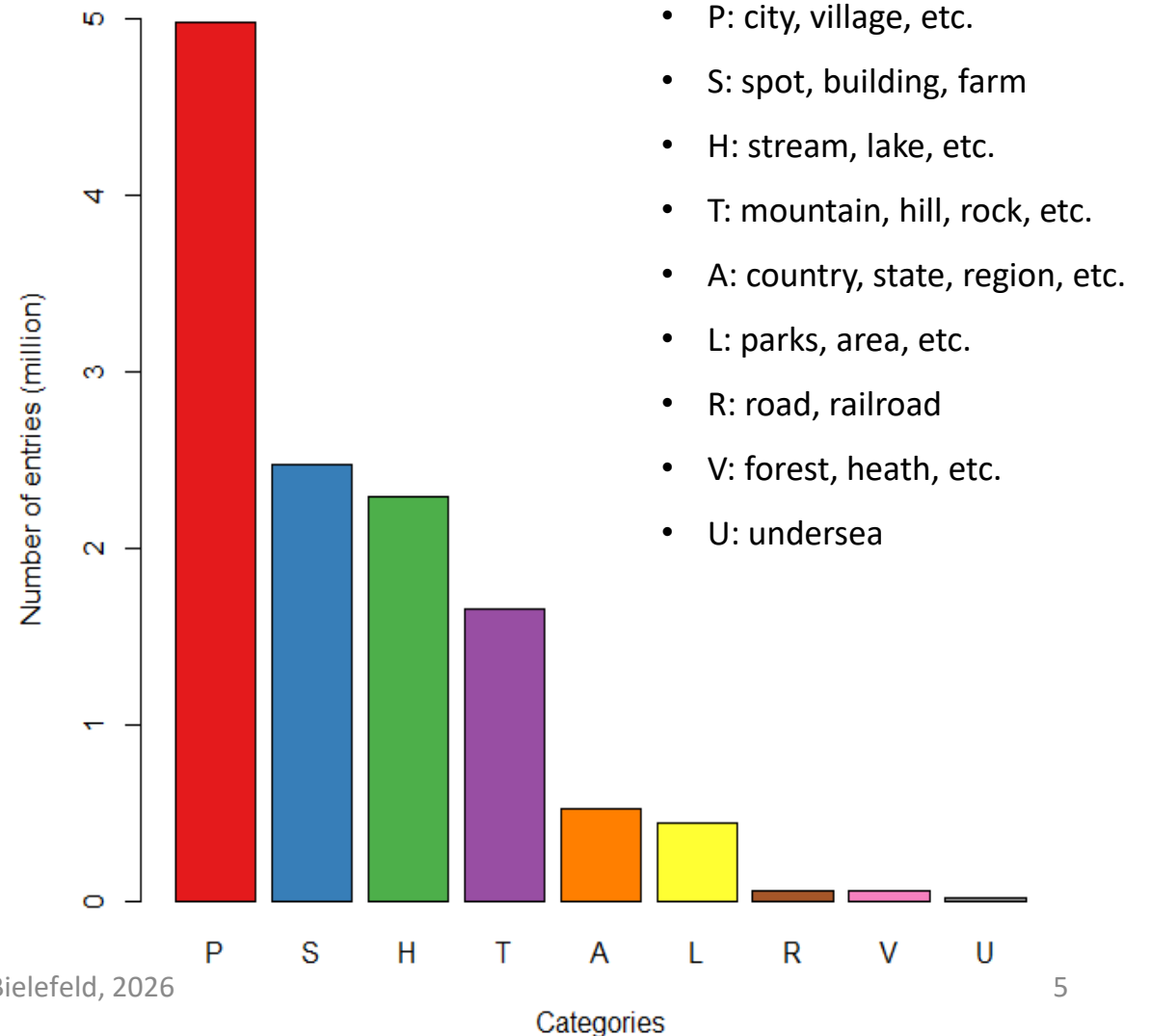
¹ Perono Cacciafoco, Francesco & Francesco Cavallaro. 2023. *Place Names: Approaches and perspectives in toponymy and toponomastics*. Cambridge: Cambridge University Press.

Approaches to toponomastics

- *Particularistic*: a focus on individual meanings and the history they encapsulate; archival work uncovering earlier recorded versions of the name
 - E.g. Eichler, Ernst and Hans Walther. 2001. *Historisches Ortsnamenbuch von Sachsen*, vol. 1–3. Berlin: Akademie Verlag.
- *Pattern-seeking*: a focus on the more frequently occurring elements of place names, including generic elements; distributional maps
 - E.g. Campbell, Jon C. 1991. Stream generic terms as indicators of historical settlement patterns. *Names* 39(4): 333–366.
- NB: The approaches are complementary, not mutually exclusive
- Note: Terminology is ours, but cf. Tent (2015) for a similar observation and related terms

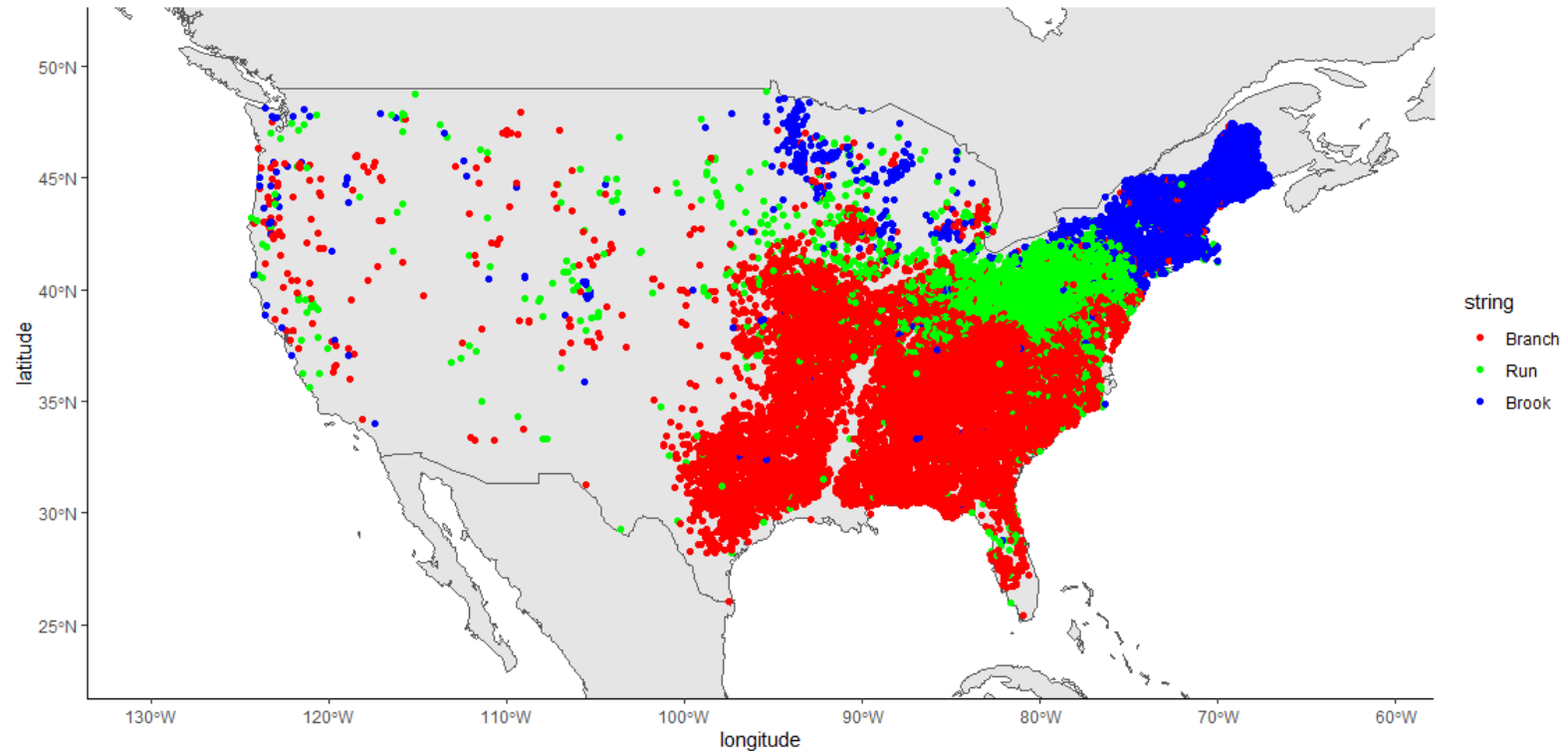
GeoNames: A toponomastic resource

- <https://www.geonames.org/>
- an evolving database of public domain geographical data, aggregated from 400+ sources
- ~12.5 million entries
- 253 countries
- 9 feature classes



Features of our package

- Generate maps using data from [geonames.org](https://www.geonames.org)
- Find frequent toponyms
 - in countries
 - in user-specified areas
 - in user-specified areas relative to the rest of a country
- Compare frequency between areas
 - simple statistical analysis



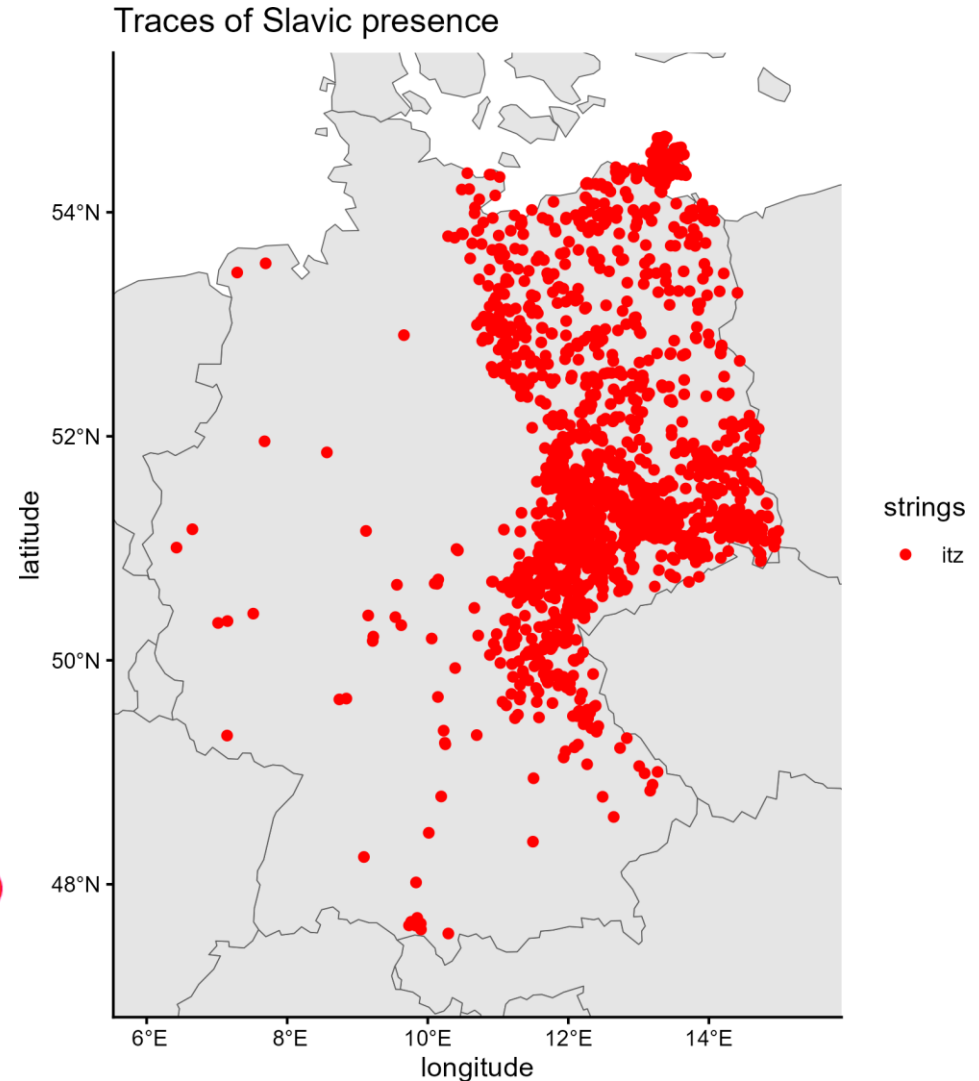
Example of hydronymic generic terms in USA (minus Alaska and Hawaii)

Example: Traces of Slavic presence

At the end: Link to Github with script for all following examples.

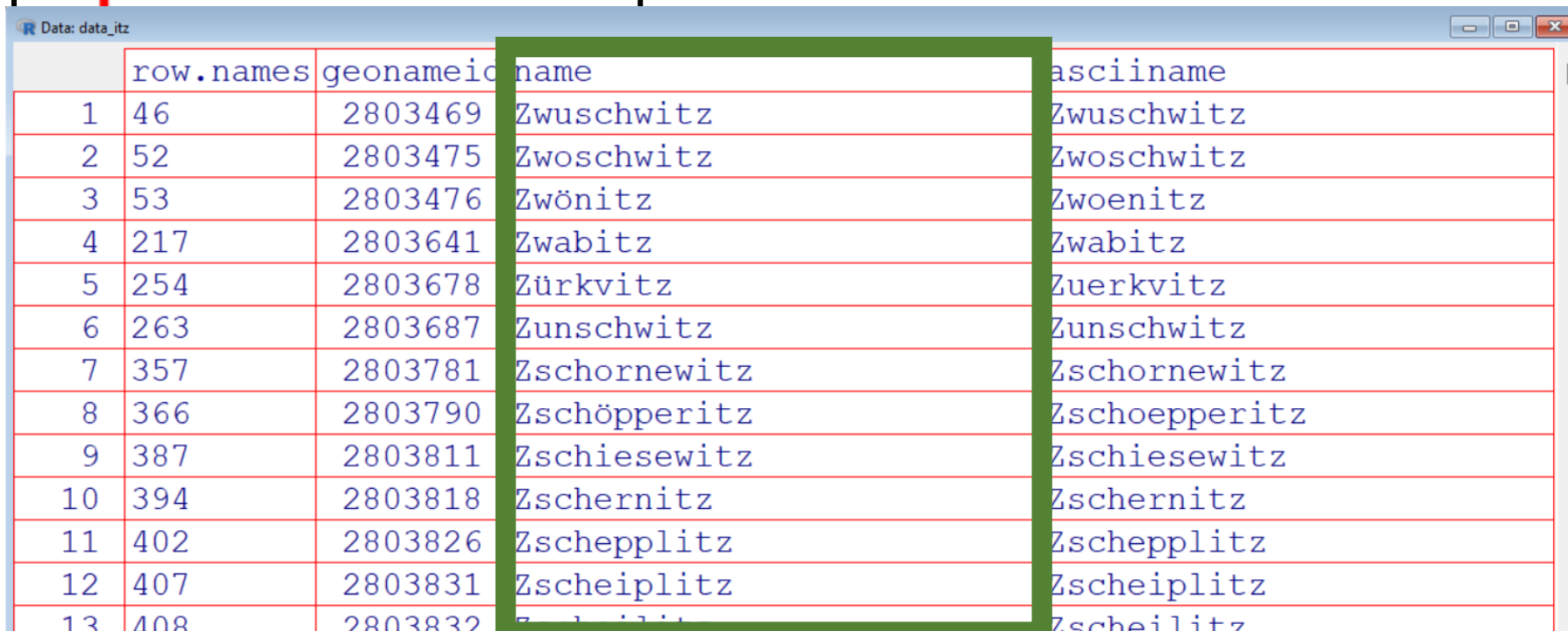
- The ending *-itz* (< *-ici* or *-ica*) means 'place' in Slavic languages
- Its occurrence in Germany indicates where Slavic presence was concentrated

```
> data_itz <- top(strings = "itz$", countries = "DE")  
> mapper(data_itz, title = "Traces of Slavic presence")  
> |
```



Data frame structure: filtering of GeoNames

```
> View(data_itz)
```



	row.names	geonameid	name	asciiname
1	46	2803469	Zwuschwitz	Zwuschwitz
2	52	2803475	Zwoschwitz	Zwoschwitz
3	53	2803476	Zwönitz	Zwoenitz
4	217	2803641	Zwabitz	Zwabitz
5	254	2803678	Zürkvitz	Zuerkvitz
6	263	2803687	Zunschwitz	Zunschwitz
7	357	2803781	Zschornewitz	Zschornewitz
8	366	2803790	Zschöpperitz	Zschoepperitz
9	387	2803811	Zschie sewitz	Zschie sewitz
10	394	2803818	Zschernitz	Zschernitz
11	402	2803826	Zschepplitz	Zschepplitz
12	407	2803831	Zscheiplitz	Zscheiplitz
13	408	2803832	Zscheilitz	Zscheilitz

- By default, the filter (regular expression given in 'strings') applies to the **'name'** column
- Further columns after 'asciiname' specify, i.a., the admin. unit

Data frame structure: filtering of GeoNames

```
> colnames(data_itz)
[1] "geonameid"      "name"
[3] "asciiname"      "alternatenames"
[5] "latitude"       "longitude"
[7] "feature class"  "feature code"
[9] "country code"   "cc2"
[11] "admin1 code"    "admin2 code"
[13] "admin3 code"    "admin4 code"
[15] "population"     "elevation"
[17] "dem"            "timezone"
[19] "modification date" "group"
```

- By default, the filter (regular expression given in 'strings') applies to the **'name'** column
- Further columns after 'asciiname' specify, i.a., the admin. unit

Frequent toponyms

- Most frequent six-character endings in Brazil:

```
> topFreq(countries = "BR", len = 6, limit = 10)
```

```
toponyms
```

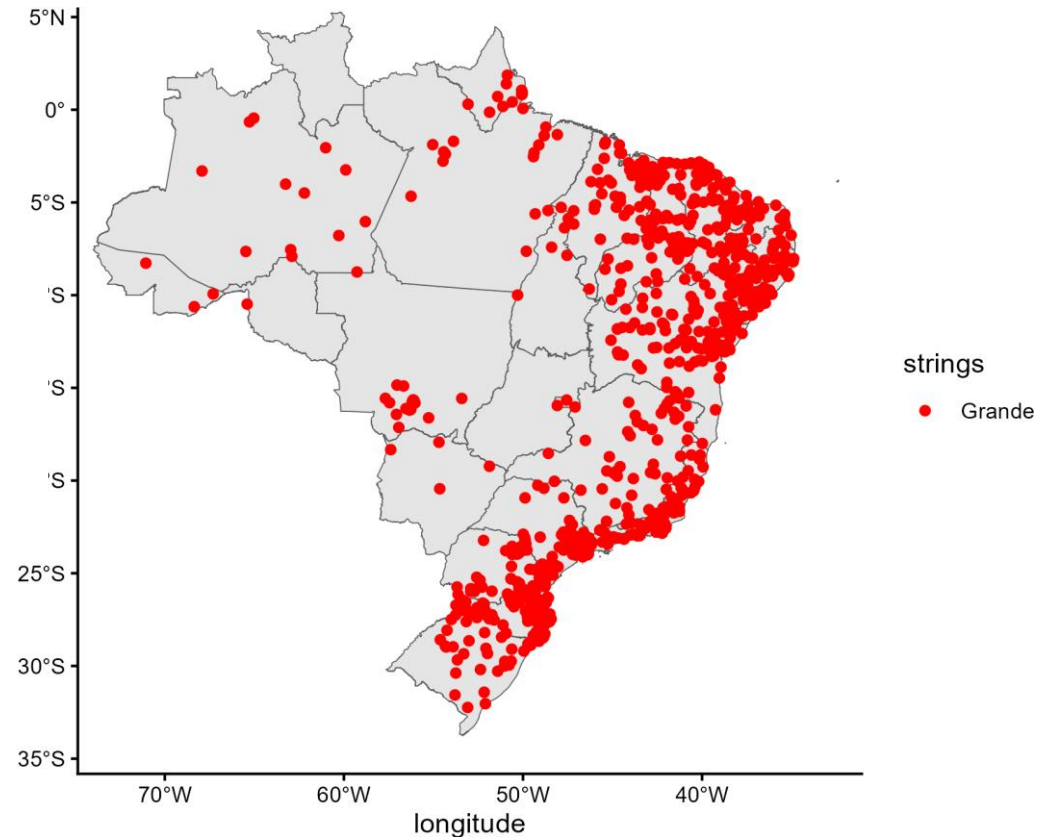
Grande\$	Vista\$	Alegre\$	ntônio\$	o José\$	erança\$	a Nova\$	o João\$
874	798	538	529	507	380	379	362
Pedro\$	lândia\$						
338	325						



```
> grande <- top(strings = "Grande$", countries = "BR")
```

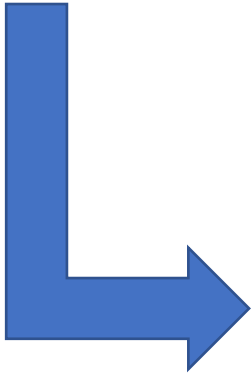
```
> mapper(grande, regions = 1)
```

```
> |
```



Further orientation

> ?top



top {toponym}

R Documentation

Selection of Toponyms

Description

This function returns coordinates of selected toponyms (strings).

Usage

top(strings, countries, ...)

Arguments

strings

character string vector with regular expressions to filter data.

countries

character string vector with country designations (names or ISO-codes).

...

Additional parameters:

- `feat.class` character string vector. Selects data only of those feature classes (check <http://download.geonames.org/export/dump/readme.txt> for the list of all feature classes). By default, it is `P`.

Toponymic studies in India

- the *Aṣṭādhyāyī* by Pāṇini contains information on place names (cf. Agrawal 1953)
- e.g., Place Names Society of India in Mysuru:
 - Journal: *Bhāratīya Sthalanāma Patrikā / Studies in Indian Place Names*
 - only few works study large distributions & show maps (such as Acharya 2019)
- see also *Mapping Place Names of India* by Anu Kapur (2019)
... and more

Examples from India: Konkan region



https://commons.wikimedia.org/wiki/File:Konkan_Districts.png

Examples from India: Konkan region

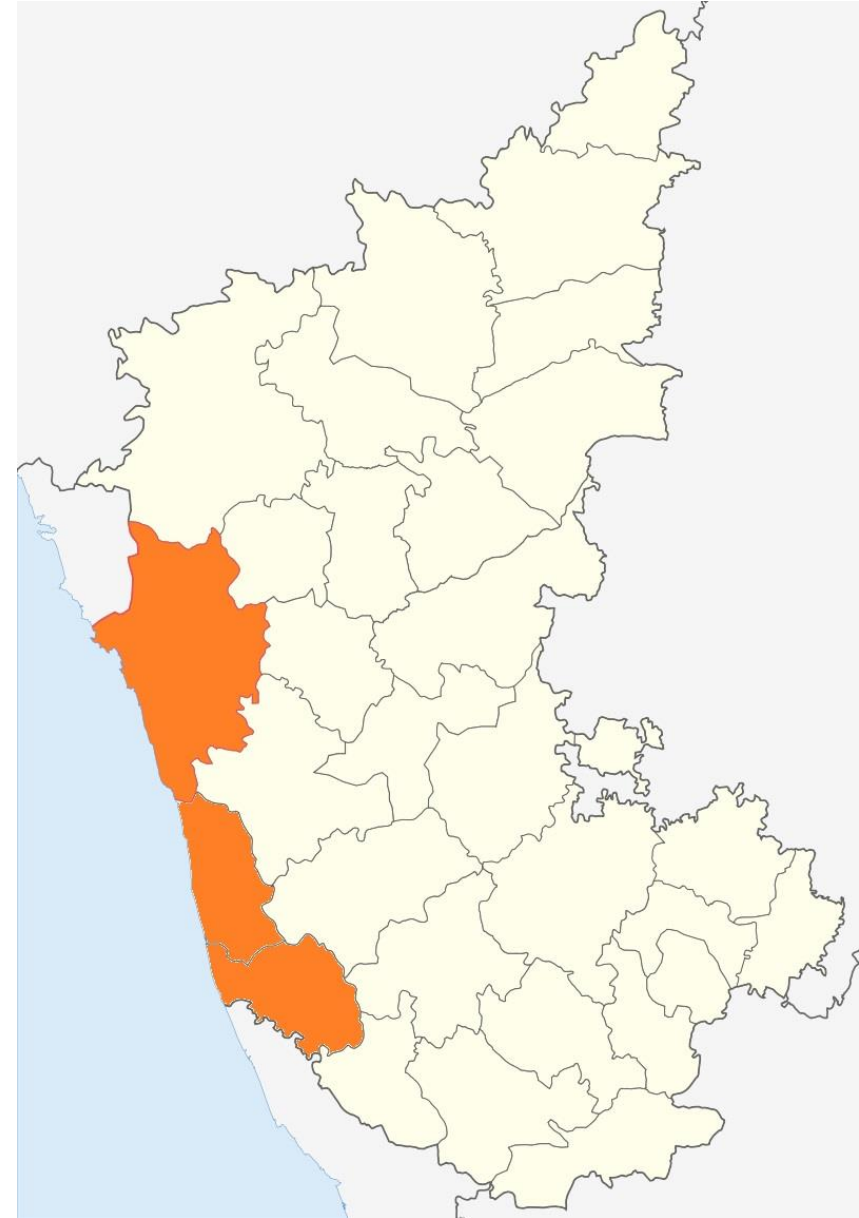


https://commons.wikimedia.org/wiki/File:Konkan_Districts.png

Konkan region



https://commons.wikimedia.org/wiki/File:Konkan_Districts.png



<https://commons.wikimedia.org/wiki/File:Canara.jpg>

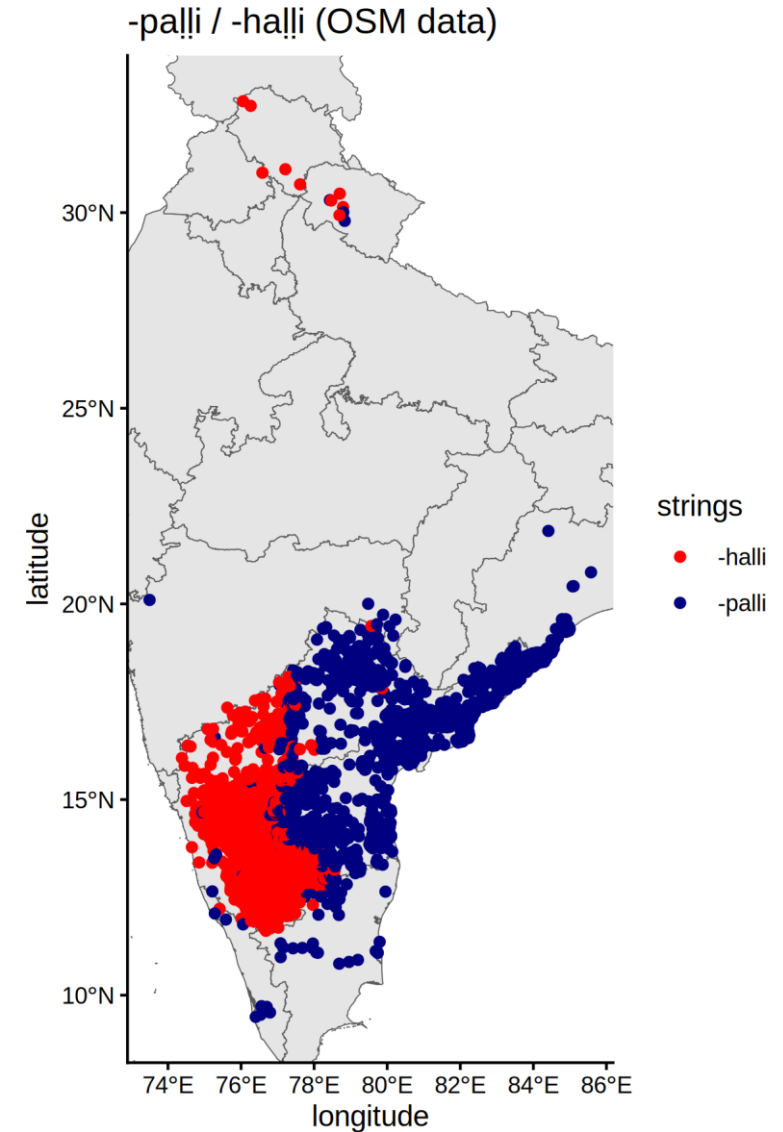
Konkan region

Southworth (2005): Maps possible (!) Dravidian place name elements in Maharashtra, focus on coastal region

- identification and description of generic elements but no maps:
 - *Toponymy of Canara* by Severine Silva (1963)
 - *Place-Names of Southern India: A Generic Approach to Toponymy* by Karimpumannil Mathai George (1986)
- *Dravidian Etymological Dictionary* by Thomas Burrow & Murray Barnson Emeneau (1984)
- **Let's map some of these names using toponym**

paḷli / *haḷli* ‘hamlet; village etc.’

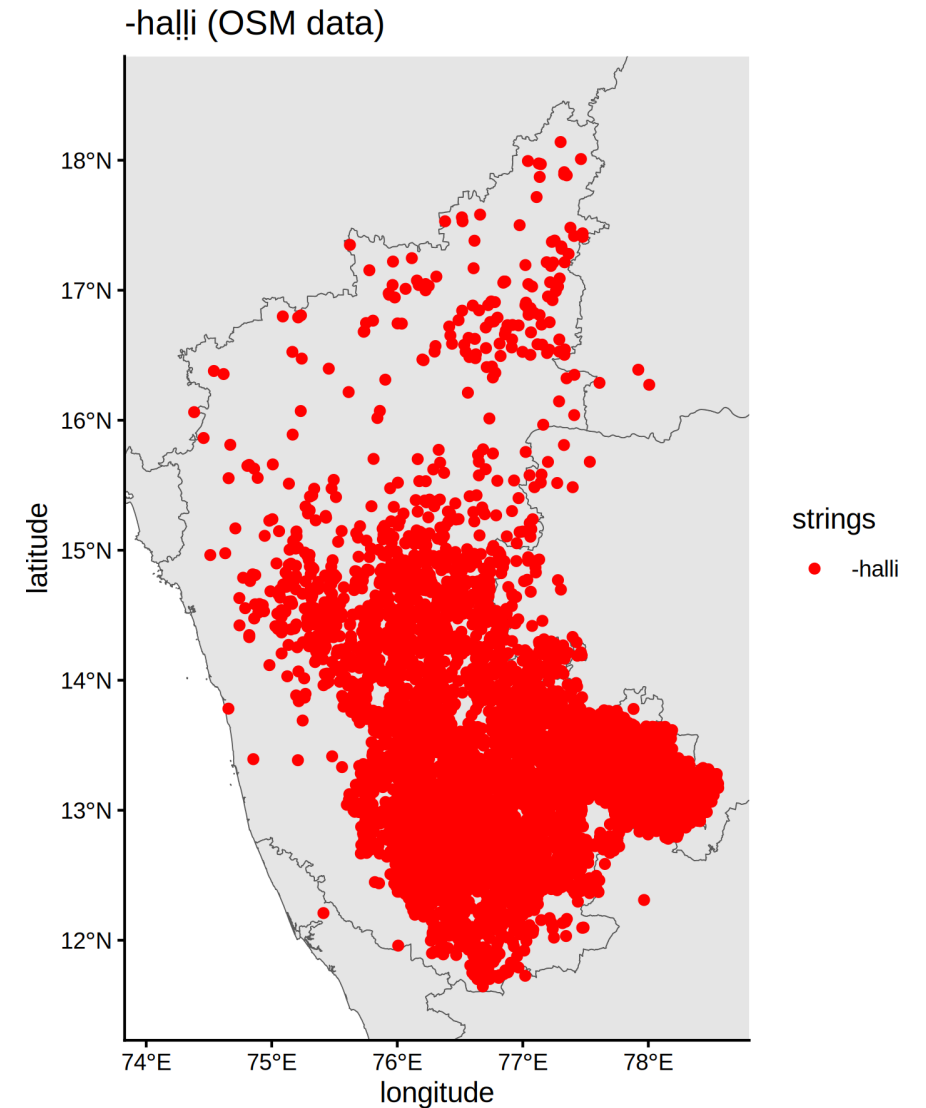
- attested for Tamil, Telugu, Malayalam, Kannada (George 1986: 203–205)
- several variants and differing meanings
 - e.g. also ‘temple’ (Burrow & Emeneau 1984, entry 4018)
- custom data used, here:
 - Open Street Map data
- retroflex laterals are not distinguished:
 - *palli* & *paḷli* etc. collapsed



paḷli / *haḷli* ‘hamlet; village etc.’

I restricted the map to:

- variant *haḷli*
- and Karnataka & surroundings
- word-initially: * $p > h$
 - from 10th century on in Kannada (Krishnamurti 2003: 120)
- few occurrences in the north
- very few in the coastal region
- **Which toponyms are common there?**



mane 'house; dwelling etc.'

Burrow & Emeneau (1983: 424, entry 4776):

Tamil:

maṇai 'house; dwelling; family, etc.'

Malayalam:

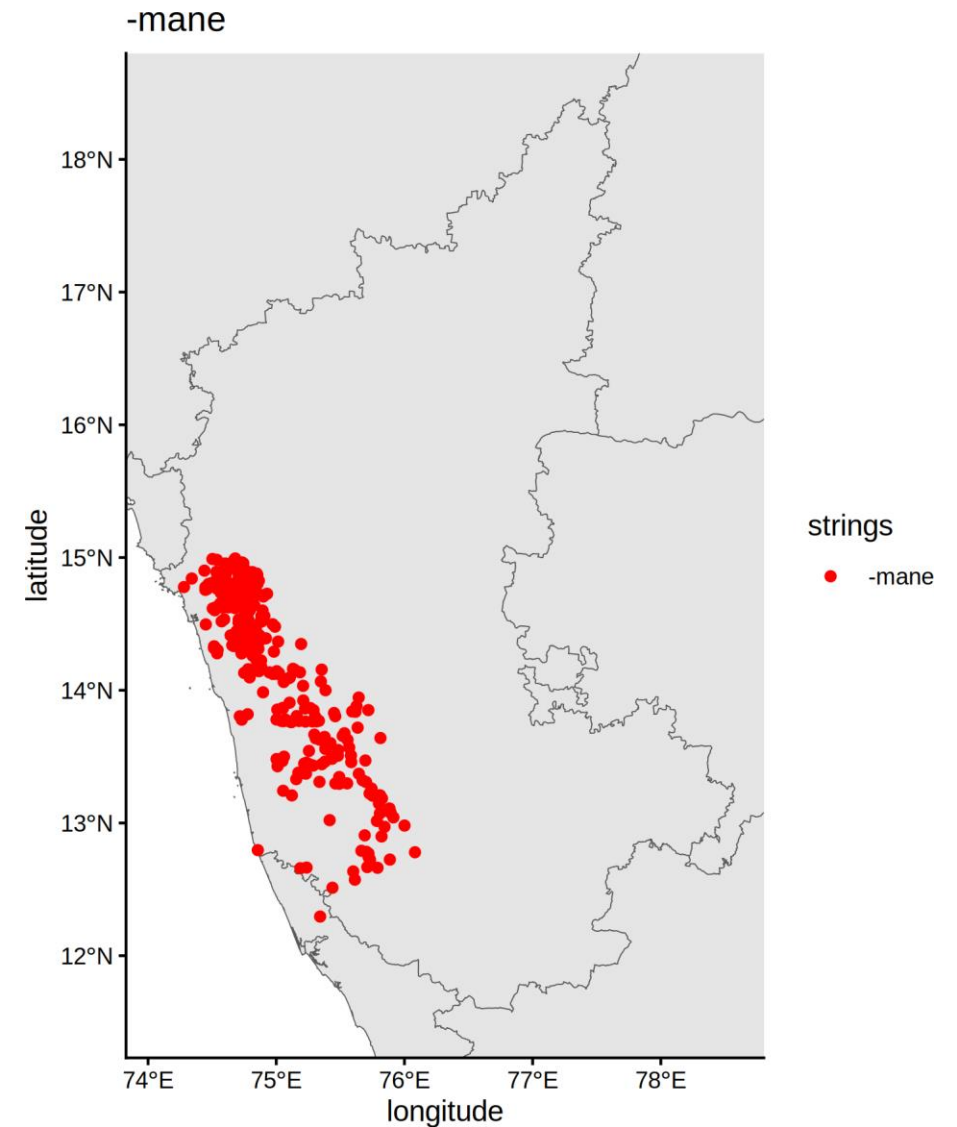
mana 'house; etc.'

Kannada:

mane 'habitation; abode; house etc.'

➤ possible variant: *mani* not frequent at the west coast

(see also George 1986: 235, no. 189)



gadde ‘field, esp. paddy field’

Burrow & Emeneau (1983: 126, entry 1355):

Tamil:

kaṛaṇi ‘field, paddy field, mud’

Malayalam:

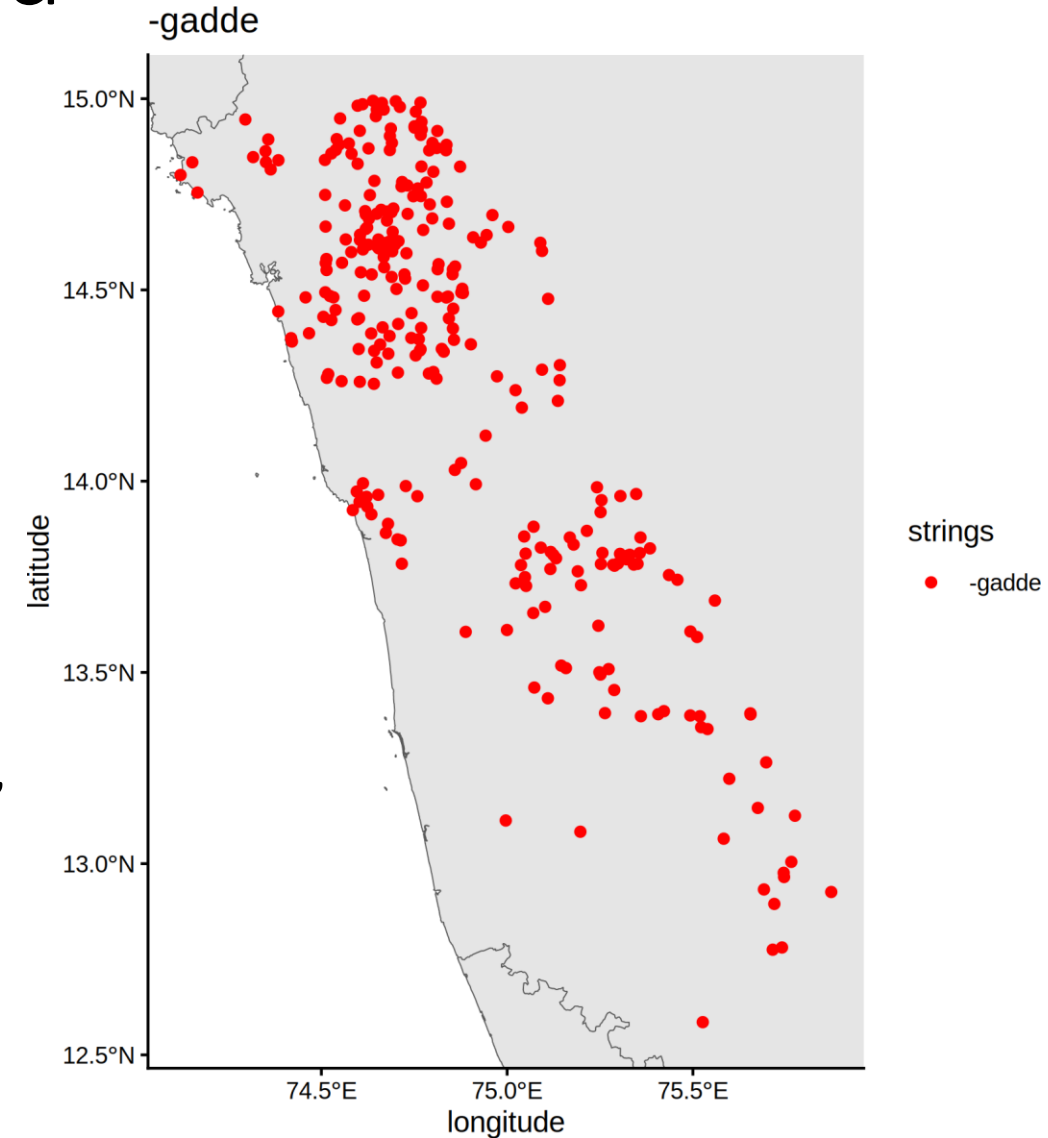
kaṛani ‘mire, rice field’

Kannada:

gadde, garde ‘field, esp. paddy field’

gaṇṇa ‘a division of paddy-land, a paddy field’

(see also George 1986: 118, no. 51)



bail(u) ‘open space or field etc.’

Burrow & Emeneau (1983: 474, entry 5258):

Tamil:

vayal, vayakkal ‘paddy field; open space etc.’

Malayalam:

vayal, vail ‘paddy field’

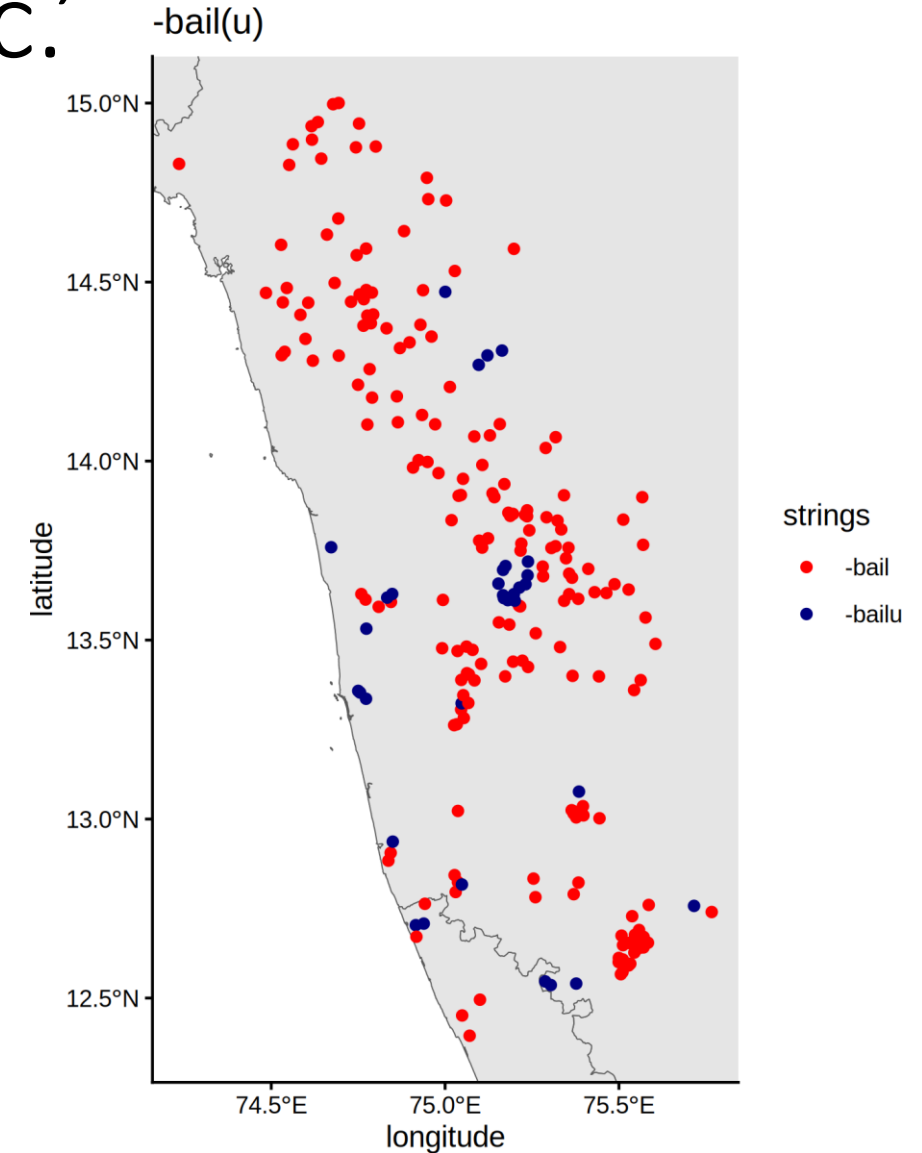
Telugu:

bayalu ‘open space or field; plain; meadow’

Kannada:

bayal(u), bayil(u), baylu ‘open space, field’

(see also George 1986: 255, no. 217)



beṭṭu ‘big hill; mountain’

Burrow & Emeneau (1983: 497, entry 5474):

Tamil:

viṭam, viṭari, viṇṭu ‘mountain’

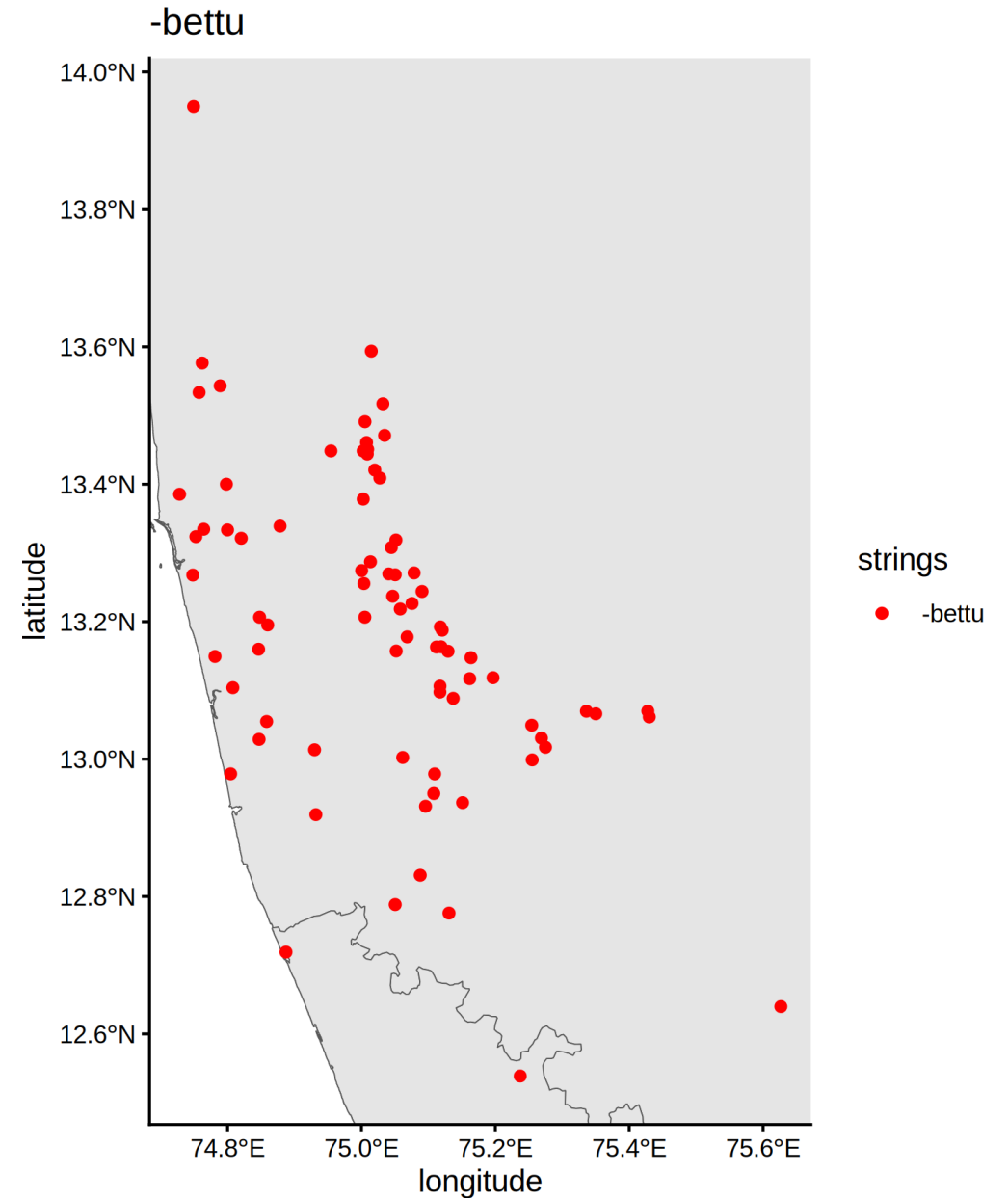
Kannada:

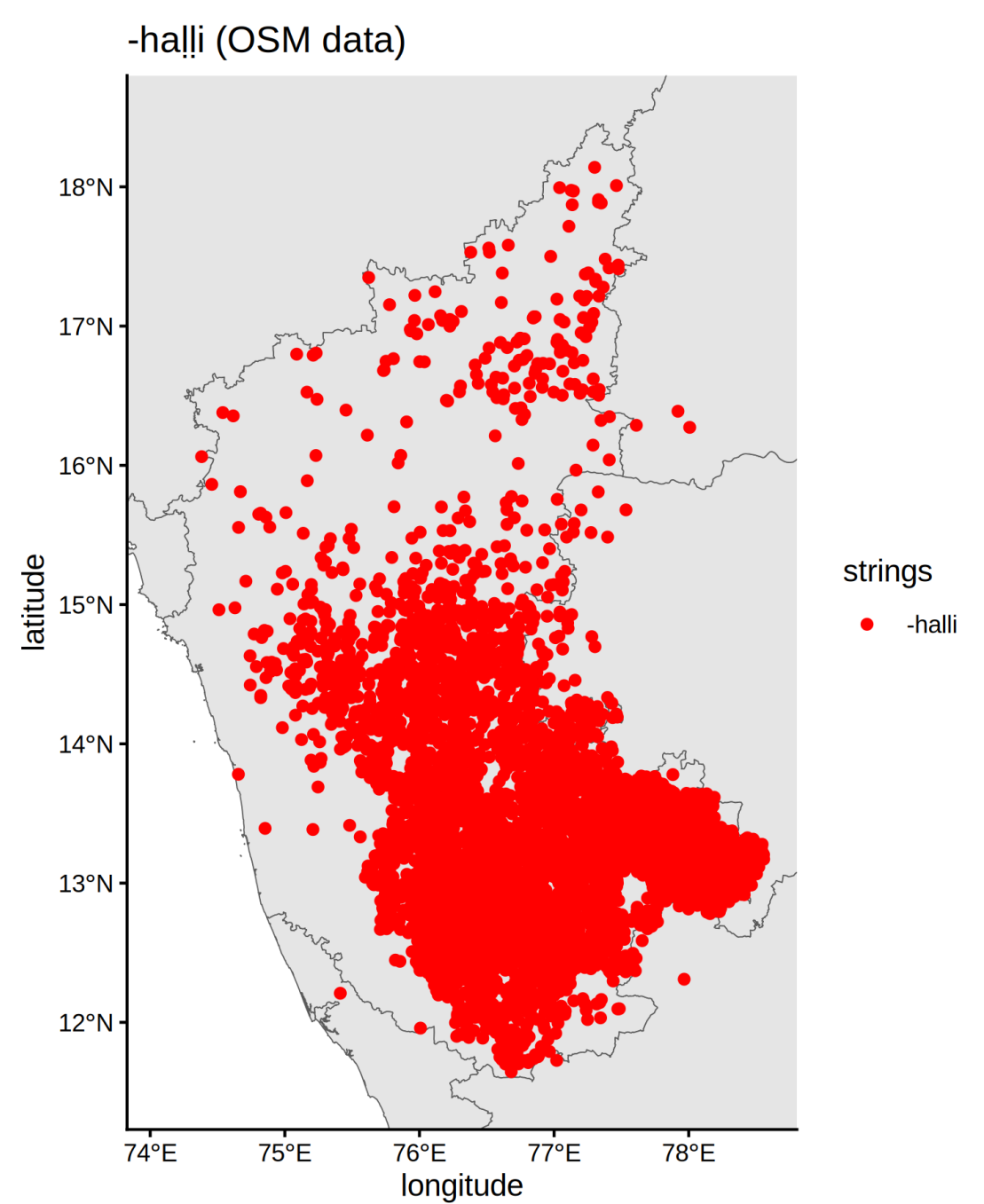
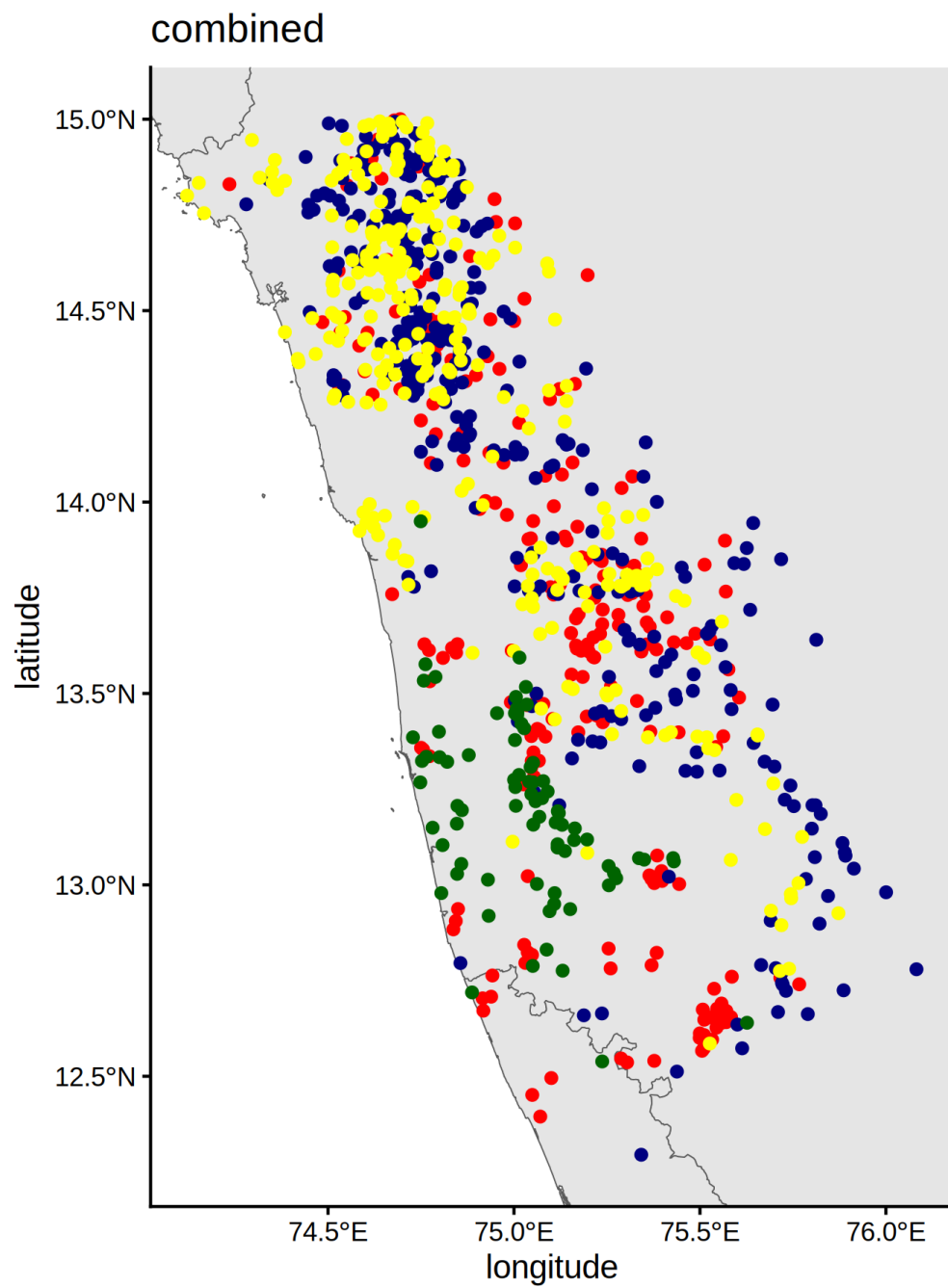
beṭṭa, beṭṭu ‘big hill; mountain’

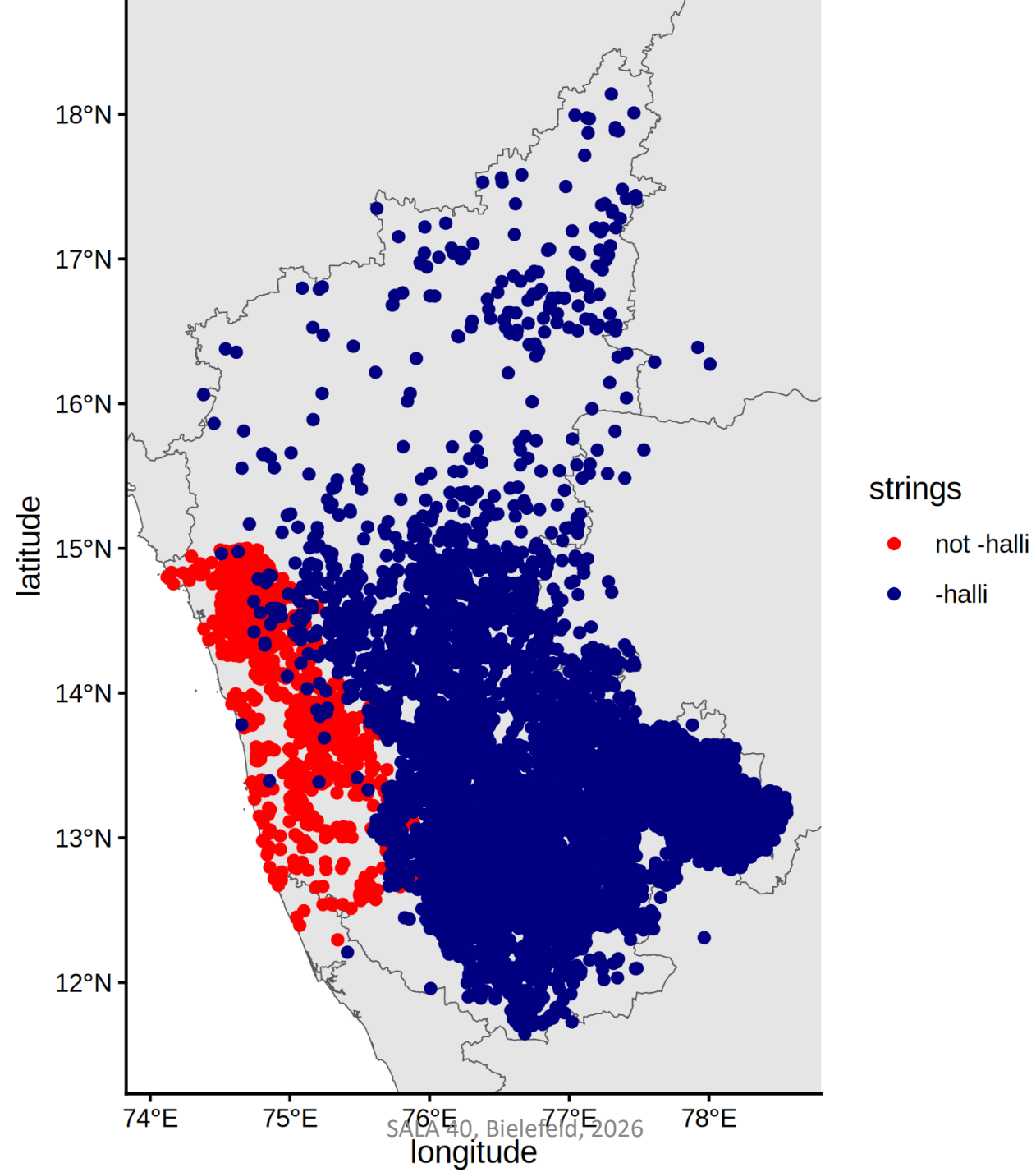
retroflex plosives are not distinguished:

beṭṭu is rendered as *bettu* in the data

(see also George 1986: 225, no. 178)







Conclusions

- `toponym` particularly useful when doing *pattern-seeking*
- to better understand generic and specific elements
 - linguistic analysis to verify meanings and structures necessary
- GeoNames is a continuously growing resource
- limitations due to (almost) exclusive availability of contemporary orthographic shapes
- `toponym` a good companion for evaluating past hypotheses
(see Wichmann & Chevallier 2025)

Possible future perspectives

- A large potential for tapping GeoNames for patterns in generic toponymic elements across the world
- Possible to incorporate other maps, including maps with more geographical and ecological information
- Permanent incorporation of other data sources, e.g. *OpenStreetMap* (import of custom data already possible)

How to get the package from within R (recommended):

```
> install.packages("toponym")
```

```
> library(toponym)
```

from CRAN:

CRAN Link:

<https://cran.r-project.org/package=toponym>

GitHub Link:

<https://github.com/Lennart05/toponym/tree/toponym-CRAN>

Link to script used in this presentation:

https://github.com/Lennart05/toponym_SALA_40_Bielefeld

toponym: Analyze and Visualize Toponyms

A tool to analyze and visualize toponym distributions. This package is intended as an interface to the GeoNames database. The functions make data and plots available for further analysis—either within R or in a chosen directory. Users can select the data selection to that region or compare regions with the remainder of countries. This package relies on the R package `geodata` (Wichmann & Chevallier (2025) <[doi:10.5195/names.2025.2616](https://doi.org/10.5195/names.2025.2616)>).

Version: 2.0.1
Depends: R (≥ 4.1)
Imports: [geodata](#), [ggplot2](#), [grDevices](#), [sf](#), [spatstat.geom](#), [spatstat.utils](#), [stats](#), [terra](#), [utils](#)
Published: 2026-04-13
DOI: [10.32614/CRAN.package.toponym](https://doi.org/10.32614/CRAN.package.toponym)
Author: Lennart Chevallier  [aut, cre], Søren Wichmann  [aut]
Maintainer: Lennart Chevallier <mail@lchevallier.de>
BugReports: <https://github.com/Lennart05/toponym/issues>
License: [GPL \(≥ 3\)](#)
URL: <https://github.com/Lennart05/toponym>
NeedsCompilation: no
Citation: [toponym citation info](#)
Materials: [README](#), [NEWS](#)
CRAN checks: [toponym results](#)

Documentation:

Reference manual: [toponym.html](#), [toponym.pdf](#)

Downloads:

Package source: [toponym_2.0.1.tar.gz](#)

Windows binaries: r-devel: [toponym_2.0.1.zip](#), r-release: [toponym_2.0.1.zip](#), r-oldrel: [toponym_2.0.1.zip](#)

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References

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- Wichmann, Søren and Lennart Chevallier. 2025. Mapping place names. *Names* 73(3): 1–19. DOI: doi:10.5195/names.2025.2616

Thank you!

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Version: 2.0.1
Depends: R (≥ 4.1)
Imports: [geodata](#), [ggplot2](#), [grDevices](#), [sf](#), [spatstat.geom](#), [spatstat.utils](#), [stats](#), [terra](#), [utils](#)
Published: 2026-04-13
DOI: [10.32614/CRAN.package.toponym](https://doi.org/10.32614/CRAN.package.toponym)
Author: Lennart Chevallier  [aut, cre], Søren Wichmann  [aut]
Maintainer: Lennart Chevallier <mail@lchevallier.de>
BugReports: <https://github.com/Lennart05/toponym/issues>
License: [GPL \(≥ 3\)](#)
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CRAN checks: [toponym results](#)

Documentation:

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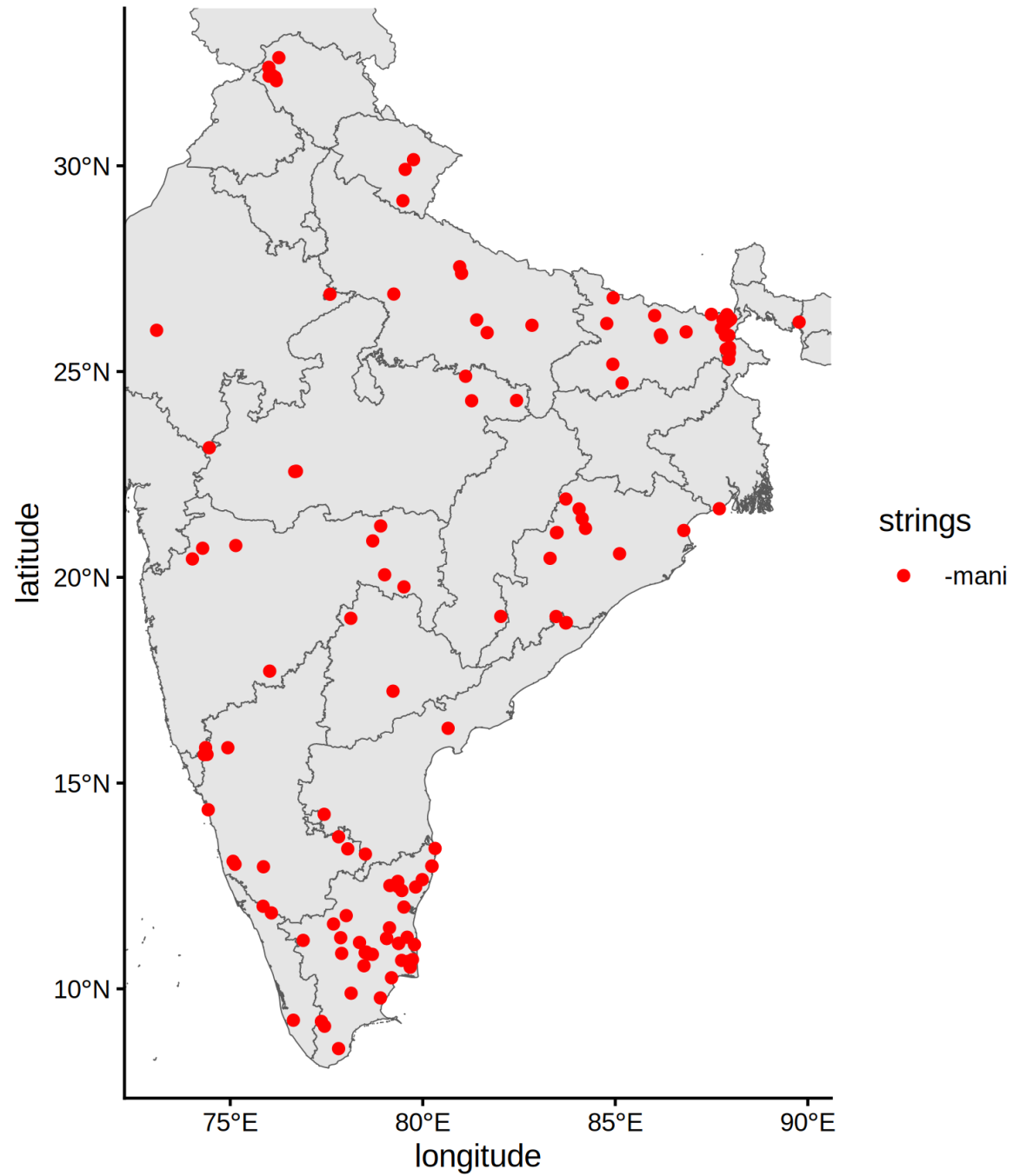
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places ending in -mani



Search for places ending in:

1. *Gāon, gāon, Gaon, gaon*
2. *Grām, grām, Gram, gram*

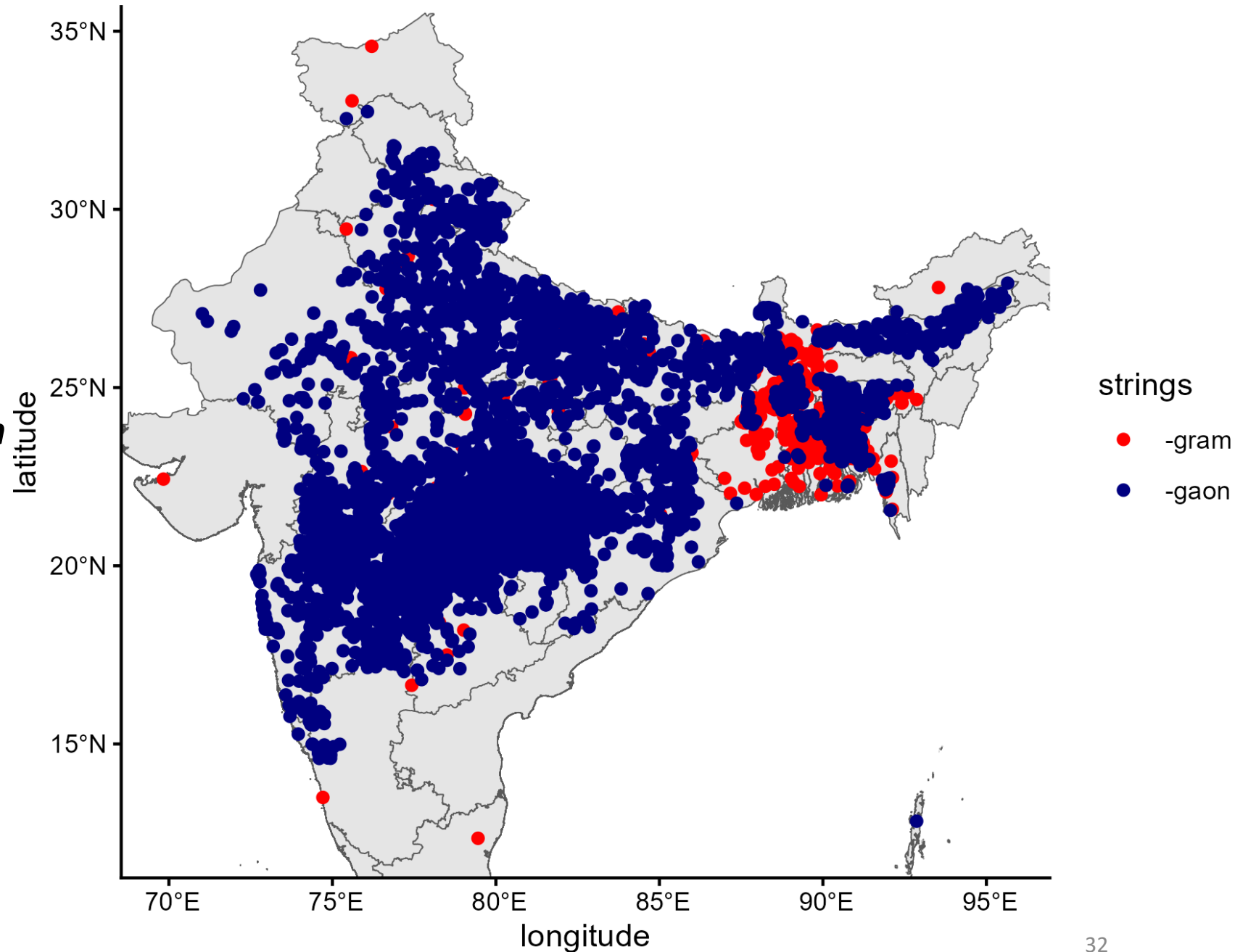
Collapsed all four variants in each group

Gāon, gāon, Gaon, gaon = **-gaon**

Grām, grām, Gram, gram = **-gram**

More variants exist.

(see also Turner (1962–1966: 235, entry 4368) on *grāma*)



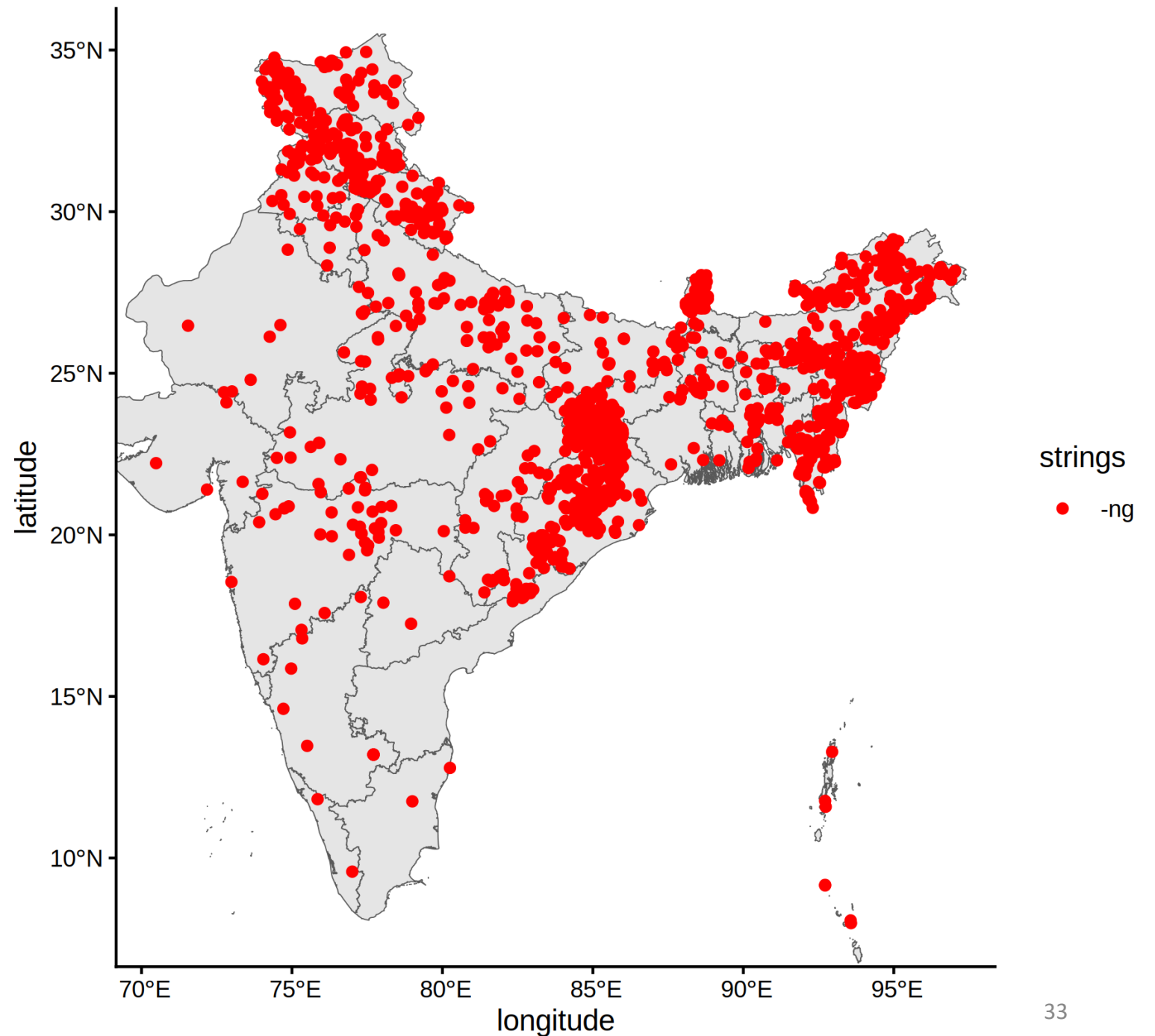
Search for places ending in:
ng

For example:
Shillong
in Meghalaya

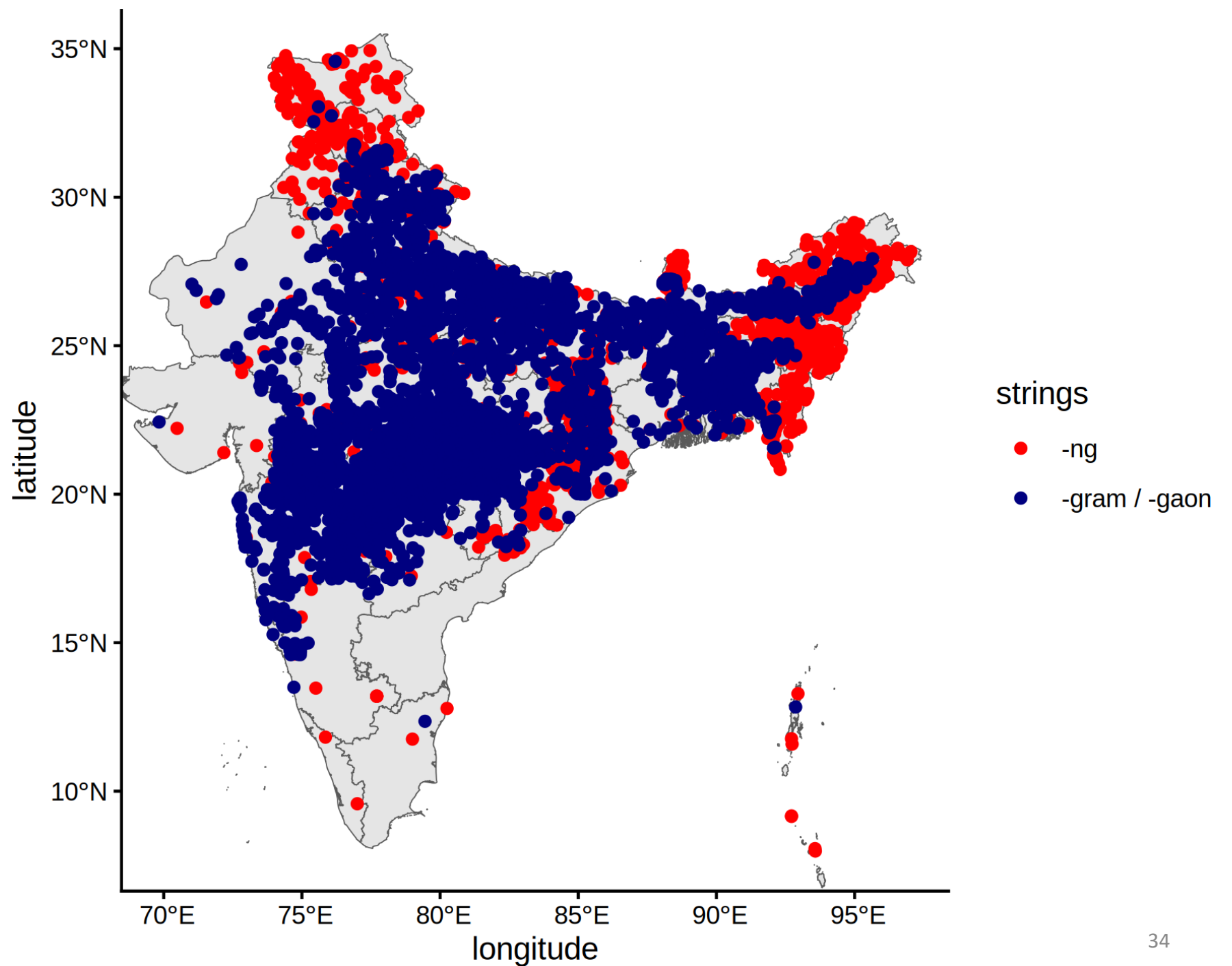
Seyochung
in Nagaland

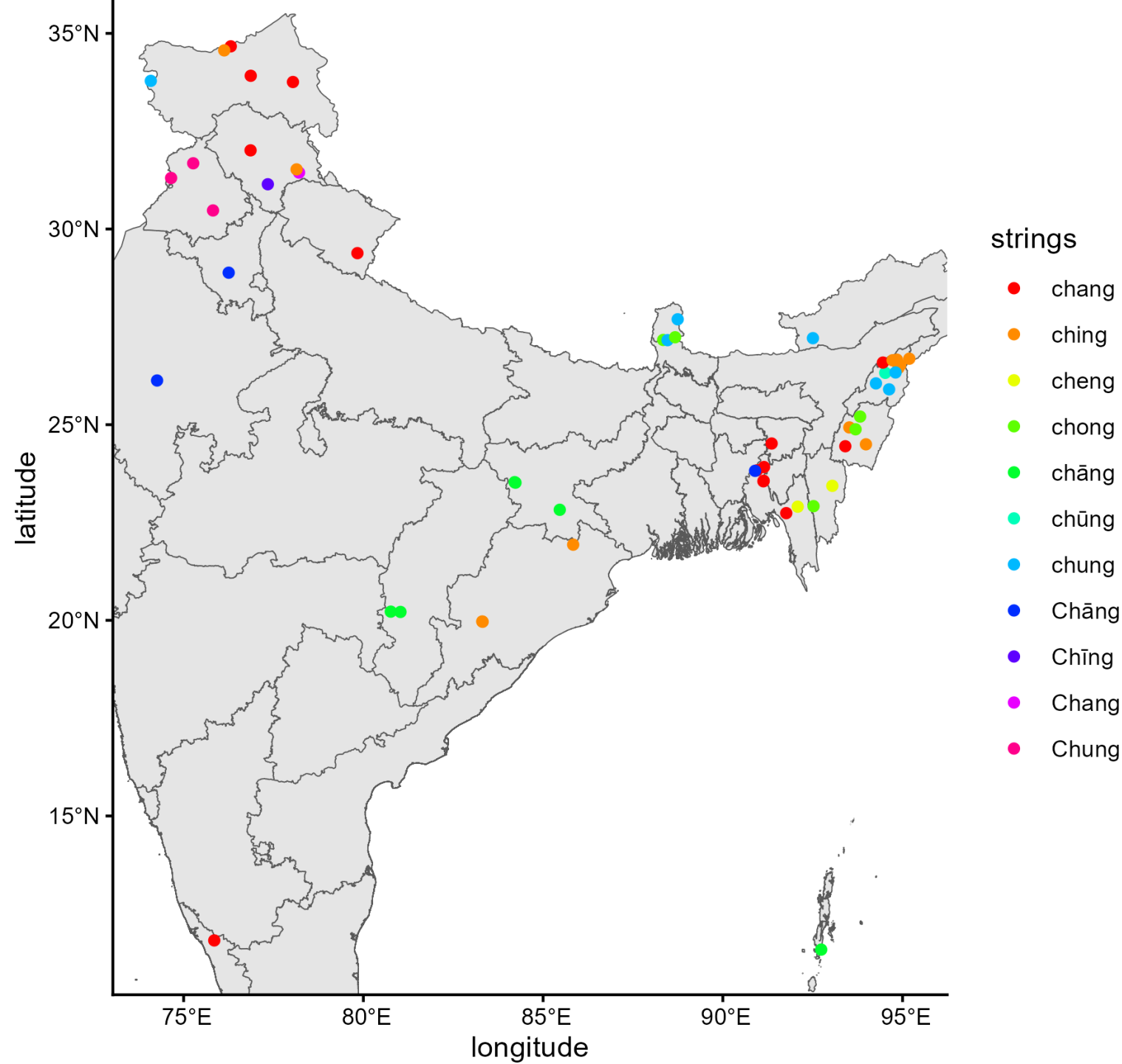
Khuangleng
in Mizoram
... etc.

Detailed work necessary!



Quick use of
package
functions reveals
different
distributions of
frequent strings.





Basic commands

- Minimally a string to filter and a country must be given
- The string is written as a regular expression, i.e. a match pattern
- A number of other parameters are possible, e.g. feature class
- Countries are given as names (e.g. 'Italy' and 'Switzerland') or ISO-codes
- `c ( )` combines strings to a vector
- A data frame is returned into the console or can be saved as object (e.g. `data sopra`)

```
> data_sopra <- top(strings = "Sopra$", countries = c("IT", "CH"))  
> |
```